

Assignment 3 Coa

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Question 1

a) $9280h = 1001\ 0010\ 1000\ 0000_2$

$903Ah = 1001\ 0000\ 0011\ 1010_2$

$$\begin{array}{r} 1001\ 0010\ 1000\ 0000 \\ 1001\ 0000\ 0011\ 1010 \\ \hline 1\ 0010\ 0010\ 1011\ 1010 \end{array}$$

 ↑
 Carry Out

$0010\ 0010\ 1011\ 1010_2 = 22BAh$

$ZF = 0$; since the result of 16 bit is not equal to zero

$SF = 0$; MSB is 0 ... positive number

$CF = 1$; carry out is 1

$OF = 1$; $Cin(1) + Cout(0) = 1$

ZF	SF	CF	OF	AF	PF
0	0	1	1	0	0

b) $8743h = \underbrace{1000}_{4\text{ bit opcode}}\ \underbrace{0111\ 0100\ 0011}_{12\text{-bit operand}}$

clock	IP/PC	MAR	MBR	IR	BX	COA	Micro - Operation
t_0	409				0003	0005	IP/PC = 409
t_1	40A	40A			0003	0005	MAR ← PC
t_2	40B	40A	8743		0003	0005	MBR ← M[MAR], PC ← PC + 1
t_3	40B	40A	8743	8743	0003	0005	IR ← MBR
t_4	40B	743	8743	8743	0003	0005	MAR ← IR[11:0]
t_5	40B	743	0005	8743	0003	0005	MBR ← M[MAR]
t_6	40B	743	0008	8743	0003	0005	MBR ← MBR + BX
t_7	40B	743	0008	8743	0003	0008	M[MAR] ← MBR

2(a)

$$(i) t_n = 57 + 25 + 45 + 35 + 45 + 40 \\ = 247 \text{ ns}$$

Total time for 1500 instructions ($T_{\text{without pipeline}}$)

$$T_{\text{without pipeline}} = \cancel{t_n} \times N$$

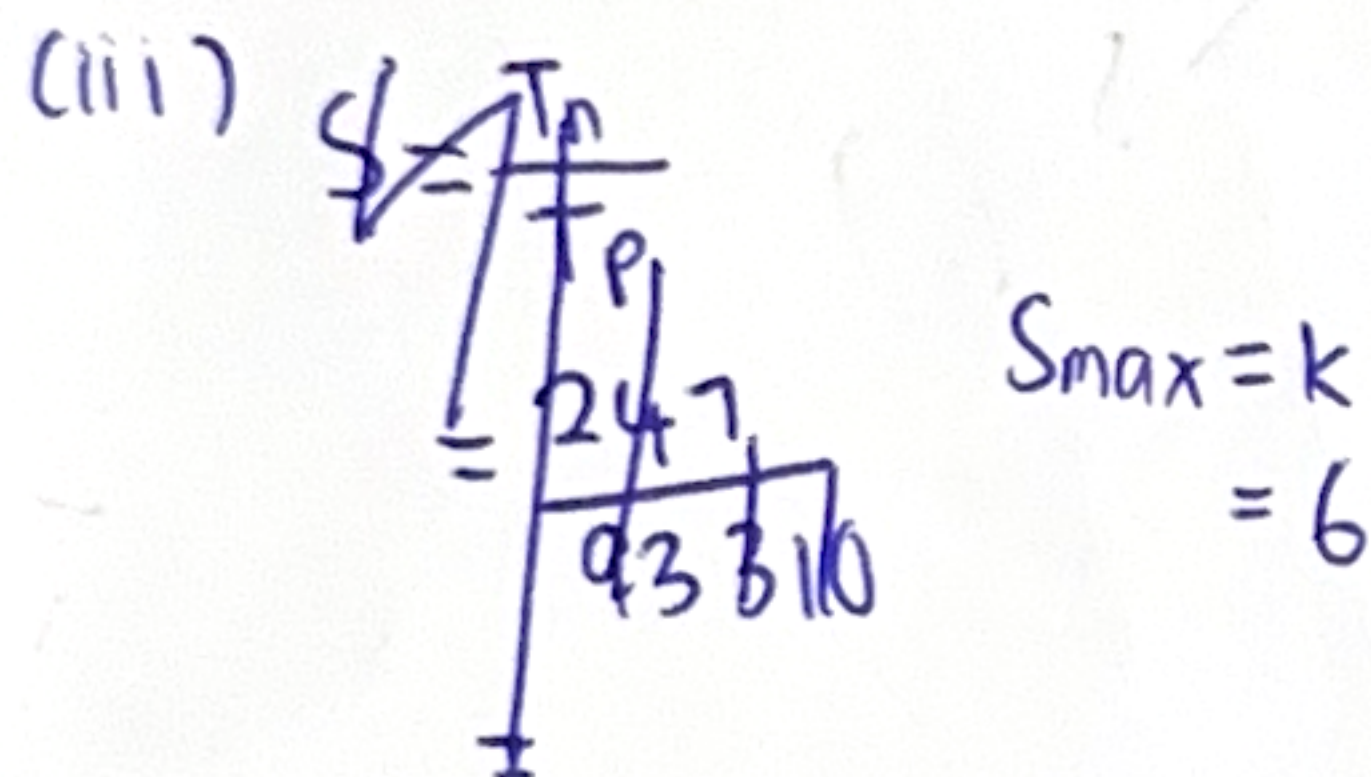
$N = 1500$ instructions

$$\cancel{T_{\text{without pipeline}}} = 1500 \times 247 \\ T_n = 370500 \text{ ns}$$

$$(ii) \# T_p = t_p(k + (n-1))$$

$$t_p = 57 + 5 \\ = 62 \text{ ns}$$

$$T_p = 62(6 + (1500 - 1)) \\ = 62(1505) \\ = 93310 \text{ ns}$$



$$(iv) S = \frac{T_n}{T_p} \\ = \frac{370500}{93310} \\ = 3.971$$

(b)

(i)

Instruction	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
L1: CMP	F1	D1	F0	E1	S0						
L2: ADD		F1	D1	F0	E1	S0					
L3: JMP			F1	D1	F0	E1	S0				
L4: ADD				F1	D1	F0	X	X	X		
L5: SUB					F1	F1	D1	F0	E1	S0	

(ii)

Instruction	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
L1: CMP	F1	D1	F0	E1	S0						
L2: ADD		F1	D1	F0	E1	S0					
L3: JMP			F1	D1	F0	E1	S0				
L4 : NOP				F1	D1	F0	E1	S0			
L5: SUB						F1	D1	F0	E1	S0	